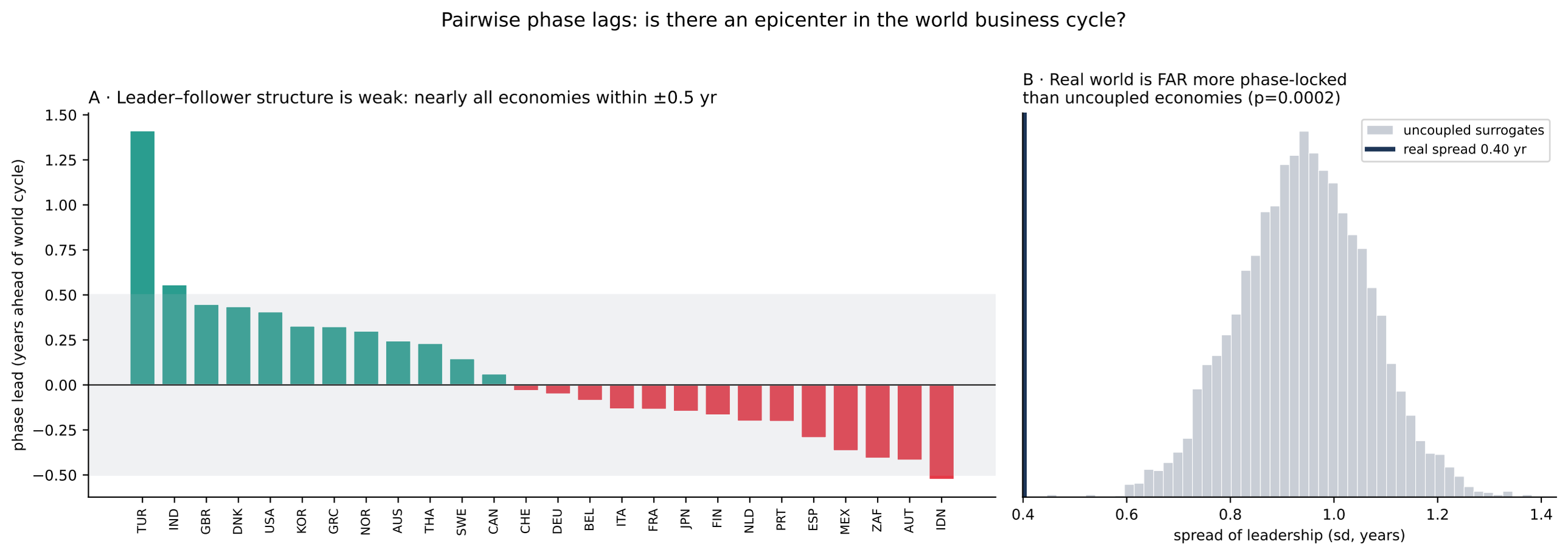
Pairwise phase lags: is there an epicenter in the world business cycle?
A · Leader–follower structure is weak: nearly all economies within ±0.5 yr
phase lead (years ahead of world cycle)
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
TUR
IND
GBR
DNK
USA
KOR
GRC
NOR
AUS
THA
SWE
CAN
CHE
DEU
BEL
ITA
FRA
JPN
FIN
NLD
PRT
ESP
MEX
ZAF
AUT
IDN
B · Real world is FAR more phase-locked
than uncoupled economies (p=0.0002)
uncoupled surrogates
real spread 0.40 yr
0.4
0.6
0.8
1.0
1.2
1.4
spread of leadership (sd, years)